AI-BASED EYE DISEASE IDENTIFICATION & HEALTH RECOMMENDATION SYSTEM

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CLINICAL OPHTHALMOLOGY & EXPLAINABLE AI DIAGNOSTIC REPORT

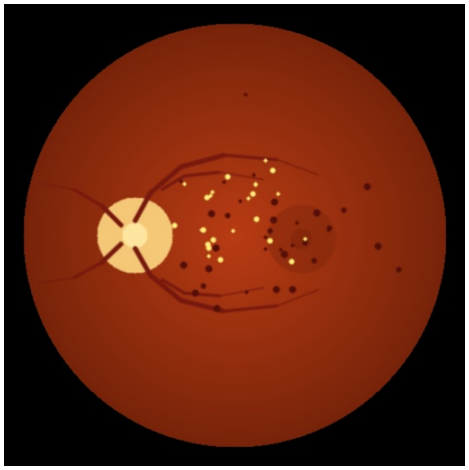
Patient Name: Test Patient	**Patient ID:** PAT-TEST-01	**Exam Date:** 23 Sep 2026
Age / Gender: 45 yrs / Male	**Contact:** 1234567890	**Medical History:** None

1. ARTIFICIAL INTELLIGENCE DIAGNOSTIC SUMMARY

Detected Condition	Severity Staging	AI Confidence	Overall Risk Index
Diabetic Retinopathy	**Moderate**	93.0%	65.0 / 100

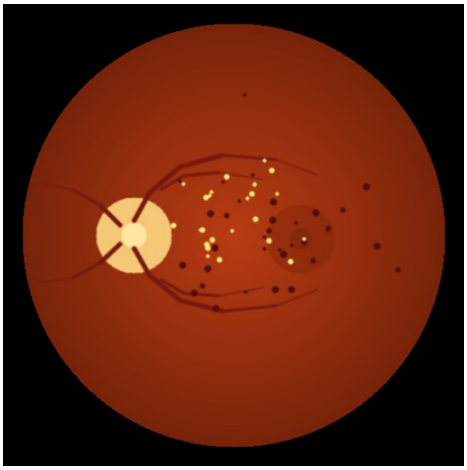
Cup-to-Disc Ratio (CDR): 0.35 **Macular Integrity:** 88.0% **Lesion Density:** 24.0% **Vessel Tortuosity:** 1.2

2. EXPLAINABLE AI (XAI) GRAD-CAM LESION LOCALIZATION



Original Retinal Fundus

Full vascular tree & optic disc



Grad-CAM Explainability Heatmap

Warm regions (Red/Yellow) highlight focal lesions

3. PERSONALIZED NUTRITIONAL & EYE CARE PRESCRIPTION

Target Nutrients:

• **Lutein & Zeaxanthin**: Filters high-energy blue wavelengths and strengthens macular pigment optical density (MPOD). (*Food sources: Kale, spinach, collard greens, egg yolks*)

• **Alpha-Lipoic Acid (ALA)**: Potent antioxidant that improves retinal insulin sensitivity and suppresses capillary leakage. (*Food sources: Spinach, broccoli, Brussels sprouts, organ meats*)

• **Omega-3 Fatty Acids (EPA/DHA)**: Decreases pathological retinal neovascularization and vascular endothelial growth factor (VEGF). (*Food sources: Wild salmon, sardines, chia seeds, flaxseed oil*)

Foods to Avoid:

• **Refined Sugars & Syrups**: Causes acute postprandial glycemic spikes that injure delicate retinal pericytes.

• **Ultra-Processed Trans Fats**: Exacerbates systemic vascular endothelial inflammation and microaneurysm leakage.

Daily Eye Care Habits (20-20-20 Rule):
• Strict adherence to the 20-20-20 rule: Every 20 minutes, gaze at an object 20 feet away for 20 seconds.
• Avoid vigorous straining or inverted yoga postures if severe proliferative changes are observed.

CLINICAL ACTION REQUIRED: Intermediate Stage - Clinical Attention Advised

Schedule a comprehensive in-person clinical exam within 2 to 4 weeks.

System Certification:

AI-assisted screening tool for clinical decision support. Final diagnostic confirmation must be validated by a registered ophthalmologist.

Consulting Ophthalmologist:

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Signature & Stamp